Diagnostic Expectation Distortion Propagation

Diagnostic Expectation in a Production Network: A TANK Model

This note sketches a tractable framework for studying inflation dynamics in a production network when firms differ in how they form expectations. In each sector, a share of firms forms rational expectations, while the remaining share forms diagnostic expectations in the sense of Bordalo, Gennaioli, and Shleifer (2018). Diagnostic firms overreact to recent forecast revisions. Because sectors are connected through input-output linkages, this overreaction changes not only the pricing decisions of diagnostic firms themselves but also the marginal costs faced by downstream firms. The key object is therefore not diagnosticity alone, or centrality alone, but the interaction between a sector's diagnostic share, its diagnosticity parameter, and its position in the production network.

Here "TANK" means two firm types, rational and diagnostic, rather than the usual two-household New Keynesian environment.

Motivation

Behavioral evidence suggests that real-world decision makers do not always form expectations as full-information rational agents. Bordalo, Gennaioli, and Shleifer (2018) introduce diagnostic expectations into economics and finance as a forward-looking form of representativeness-based overreaction. When a state becomes more likely after new information arrives, diagnostic agents overweight outcomes associated with that state. For a variable that follows a persistent process, this generates extrapolative forecasts: good news leads agents to overstate future good outcomes, and bad news leads agents to overstate future bad outcomes.

Carvalho, Eusepi, Moench, and Preston (2021) study how subjective beliefs can explain the joint dynamics of inflation and inflation expectations. Their framework is useful for thinking about expectation formation in inflation data, but it abstracts from a channel that is central in multisector economies: input-output linkages. In a production network, the price set by one sector becomes an input cost for other sectors. Hence, if upstream firms set prices using diagnostic beliefs, the resulting price distortion propagates through realized input prices. Downstream firms can be fully rational and still inherit the upstream diagnostic distortion through their marginal costs.

The project is built around this gap. A sector's contribution to inflation and inflation expectations should depend on both its behavioral type composition and its network position. Highly diagnostic but peripheral sectors may have small aggregate effects. Highly central sectors with low diagnosticity may propagate shocks strongly but not distort beliefs much. The potentially important cases are sectors with high diagnosticity, high diagnostic-firm shares, and high upstream centrality.

Research Questions

1. How do firms set prices in a networked economy with a mixture of rational and diagnostic firms?
2. How do a sector's diagnostic share ϕ_i, diagnosticity parameter θ_i, and network centrality κ_i jointly shape aggregate inflation Π_t and inflation expectations?
3. What are the real effects of diagnostic belief distortions when they propagate through input-output linkages?

Model

Sectors, Firms, and Beliefs

There are N sectors in a closed economy. In each sector i, there is a unit mass of firms indexed by $f \in [0,1]$. A share $1-\phi_i$ of firms forms rational expectations, denoted by $\mathbb{E}$, and a share ϕ_i forms diagnostic expectations, denoted by $\mathbb{D}$. The parameter $\phi_i \in [0,1]$ measures the prevalence of diagnostic firms in sector i.

For a variable W_t whose law of motion is understood by firms, diagnostic expectations are

$$\mathbb{D}_t[W_{t+\tau}] = \mathbb{E}_t[W_{t+\tau}] + \theta_i \left(\mathbb{E}_t[W_{t+\tau}] - \mathbb{E}_{t-1}[W_{t+\tau}]\right),$$

where $\theta_i \geq 0$ is the diagnosticity parameter in sector i. When $\theta_i = 0$, diagnostic and rational expectations coincide. When $\theta_i > 0$, firms extrapolate in the direction of the latest forecast revision.

Technology

Each firm in sector i produces using labor and intermediate inputs:

$$Y_{if,t} = e^{z_{it}} \zeta_i L_{if,t}^{\gamma_i} \prod_{j=1}^{N} X_{if,j,t}^{\alpha_{ij}},$$

with

$$\gamma_i + \sum_{j=1}^{N} \alpha_{ij} = \psi < 1.$$

Sectoral productivity is

$$z_{it} = \bar{z}_i + \epsilon_{it}^z,$$

where

$$\epsilon_{it}^z = \rho^z \epsilon_{i,t-1}^z + \nu_{it}^z, \qquad \nu_{it}^z \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\nu^2).$$

Timing

Price setting is subject to Calvo rigidity. Each period, a firm keeps its previous price with probability ω and resets its price with probability $1 - \omega$, where $\omega \in (0, 1)$.

1. Before current productivity z_{it} is realized, a firm that resets chooses $P_{if,t}$ based on its expectation mode. Rational firms use $\mathbb{E}$ and diagnostic firms use $\mathbb{D}_i$. A firm that does not reset sets $P_{if,t} = P_{if,t-1}$. The pricing objective for a resetting firm is expected profit,

$$\Gamma_{if,t} = (1 - \tau_i) P_{if,t} Y_{if,t} - W_t L_{if,t} - \sum_{j=1}^{N} P_{jt} X_{if,j,t},$$

where τ_i is a sectoral tax or subsidy.

2. After z_{it} is realized, firms choose labor and intermediate inputs to minimize costs subject to demand at the price in place for the period.

Sectoral Aggregation

Each sector has a CES aggregator:

$$Y_{it} = \left(\int_0^1 Y_{if,t}^{(\xi-1)/\xi} \, df \right)^{\xi/(\xi-1)}, \qquad \xi > 1.$$

The sectoral price P_{it} is set to maximize $P_{it} Y_{it} - \int_0^1 P_{if,t} Y_{if,t} \, df$. This aggregator earns zero profit, as in La'O and Tahbaz-Salehi (2022).

Household

The representative household maximizes

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s U(C_{t+s}, L_{t+s}), \qquad \beta \in (0,1),$$

with period utility

$$U(C, L) = \log C - \chi \frac{L^{1+\varphi}}{1+\varphi}, \qquad \chi > 0, \quad \varphi > 0.$$

Final consumption is Cobb-Douglas across sectoral goods:

$$C_t = \prod_{i=1}^{N} \left(\frac{C_{it}}{\rho_i} \right)^{\rho_i}, \qquad \sum_{i=1}^{N} \rho_i = 1.$$

The household budget constraint is

$$P_t C_t = \sum_{i=1}^{N} P_{it} C_{it} \leq W_t L_t + \sum_{i=1}^{N} \int_0^1 \Gamma_{if,t} \, df + T_t,$$

where tax revenues are rebated lump-sum:

$$T_t = \sum_{i=1}^{N} \int_0^1 \tau_i P_{if,t} Y_{if,t} \, df.$$

Nominal Anchor

To define inflation, the model needs a nominal anchor. Use an exogenous nominal demand process:

$$M_t = P_t C_t.$$

The log of nominal demand is exogenous, with constant growth μ and an optional AR(1) shock. This anchor pins down the aggregate price level and the nominal wage, so aggregate inflation is well defined:

$$\Pi_t = \frac{P_t}{P_{t-1}}.$$

Market Clearing

For every sector i,

$$Y_{it} = C_{it} + \sum_{k=1}^{N} \int_0^1 X_{kf,i,t} \, df,$$

where $X_{kf,i,t}$ is sector i's good used by firms in sector k. Labor clears:

$$L_t = \sum_{i=1}^{N} \int_0^1 L_{if,t} \, df.$$

Together with household optimality, firm cost minimization, price setting, aggregation, the nominal-demand rule, and government budget balance, these conditions define equilibrium.

Analytical Targets

These objects are to be derived.

1. How P_{it} changes with ϕ_i and θ_i.
2. Inflation Π_t and the inflation expectation $\mathbb{R}_t[\Pi_{t+\tau}]$, where $\mathbb{R}_t$ is the conditional realized expectation at period t of a variable at $t+\tau$.
3. A centrality measure κ_i for each sector. Compare inflation and inflation expectations across sectors through the interaction $\kappa_i \times \theta_i \times \phi_i$. This product denotes interaction, not necessarily multiplication.
4. How Calvo persistence ω changes the persistence of these objects.

Quantitative Analysis

Calibrate the model and compare real variables, inflation, and inflation expectations when a shock hits sectors with different centralities and diagnosticities. Vary ω to see how price persistence changes these responses.

Open Issues and Next Steps

1. **Existence and uniqueness.** The equilibrium with decreasing returns, heterogeneous beliefs, and Calvo rigidity is still to be characterized.
2. **Measurement of ϕ_i.** The diagnostic share is empirically difficult to observe.
3. **Policy.** Once the positive model is solved, ask whether monetary policy should put extra weight on sectors with high centrality and high diagnosticity, following La'O and Tahbaz-Salehi (2022).

References

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Carvalho, Carlos, Stefano Eusepi, Emanuel Moench, and Bruce Preston. 2021. "Anchored Inflation Expectations." Working Paper. BibTeX key: `carvalhoAnchoredInflationExpectations2021`.

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